

PRNG CODE::

```
import datetime
import hashlib
import matplotlib.pyplot as plt
import numpy as np

def datetime_based_prng(n=1000):
    prng_outputs = []
    timestamps = []

    for _ in range(n):
        now = datetime.datetime.now()
        timestamp_str = now.strftime("%Y-%m-%d %H:%M:%S.%f")

        hash_obj = hashlib.sha256(timestamp_str.encode())
        hex_digest = hash_obj.hexdigest()
        random_number = int(hex_digest[:8], 16)

        prng_outputs.append(random_number)
        timestamps.append(now)

    return timestamps, prng_outputs

timestamps, prng_outputs = datetime_based_prng(1000)

start_time = timestamps[0]
elapsed_seconds = [(ts - start_time).total_seconds() for ts in timestamps]

random_indices = np.random.choice(len(prng_outputs), size=5, replace=False)
random_values = [prng_outputs[i] for i in random_indices]
random_times = [elapsed_seconds[i] for i in random_indices]

print("Random 5 PRNG Outputs:")
for i, val in enumerate(random_values):
    print(f"{i+1}: {val}")

plt.figure(figsize=(10,6))
```

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plt.scatter(elapsed_seconds, prng_outputs, s=10, alpha=0.6, label='PRNG Outputs')

plt.scatter(random_times, random_values, color='green', s=50, label='Random 5
Outputs', edgecolors='black')

coeffs = np.polyfit(elapsed_seconds, prng_outputs, 1)
poly_eqn = np.poly1d(coeffs)
plt.plot(elapsed_seconds, poly_eqn(elapsed_seconds), color='red', label='Regression
Line')

plt.title("Timestamp vs PRNG Output Scatter Plot with Random 5 Highlights")
plt.xlabel("Elapsed Time (seconds)")
plt.ylabel("PRNG Output")
plt.legend()
plt.grid(True)
plt.show()

```

Random 5 PRNG Outputs:

1: 3547719087
 2: 4150567192
 3: 1254852686
 4: 3525037979
 5: 2342090029

